

Sottojit Roy

Full-Stack AI Engineer

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ABOUT ME

Senior software engineer with 8+ years of experience in **web development** and **AI-related projects**. I have worked with international clients and enjoy building practical and efficient digital solutions. I specialized in various stacks like **React.js**, **Python**(PyTorch/OpenAI/LangChain), **postgreSQL**, **Redis**, **Node.js**, **MongoDB**, **Java**, **Docker** and basic web stacks like **HTML**, **CSS**, **JavaScript**. Comfortable building and shipping in **AI-driven workflows** and contributing to **long-term architecture initiatives** that scale systems for faster product releases. Strong **product sense** and a natural collaborator with product, design, and AI teams - motivated by technology that creates meaningful impact, especially in **digital health and health tech**. Also, fully remote-capable.

Important: Value providing **practical** benefits to clients and maintaining **clear** communication and **credibility**.

WORK EXPERIENCE

PatSnap | Singapore (Remote) | Feb, 2024 - Mar, 2026

AI SaaS App Developer

- Led development of **AI-powered SaaS** features for innovation and patent intelligence platforms, improving **enterprise search** relevance and insight **discovery efficiency** by **35%** across large-scale IP datasets.
- Designed and implemented LLM-based document understanding pipelines for semantic search, entity extraction, and contextual recommendation systems, increasing **recommendation accuracy** and **metadata extraction quality** by **28%**.
- Built scalable backend architecture for vector-based retrieval and AI-assisted knowledge discovery systems, reducing **semantic query response latency** by **30%** through optimized PostgreSQL indexing and vector search infrastructure.
- Developed high-performance microservices in Python (**FastAPI**) and **Java**, supporting enterprise SaaS platforms handling **millions of API requests** monthly across **global** clients.
- Improved deployment reliability and release velocity by implementing **Kubernetes**, **Docker**, **AWS** (S3, EC2, RDS), and GitHub Actions **CI/CD** pipelines, reducing production **deployment time** by **50%** and improving infrastructure stability.
- Contributed to enhancements of AI-powered patent intelligence and analytics capabilities, enabling enterprise users to accelerate innovation discovery and competitive research workflows by **30%+** through **intelligent search and recommendation features**.

Carousell | Singapore (Remote) | May 2021 - Feb 2024

SaaS Web Developer – Scalability

- Developed scalable marketplace features for listing, search, and transaction flows, supporting **over 1 million active users monthly** and increasing platform transaction throughput by **25%**.
- Built and **optimized** ranking and recommendation systems using behavioral signals and metadata-driven models, improving listing click-through rates (**CTR**) by **30%** and overall user engagement by **10%**.
- Designed and **maintained** A/B testing infrastructure and feature flag systems, enabling rapid experimentation and increasing feature rollout efficiency by **40%** across core user journeys.
- Improved backend scalability with Redis caching, PostgreSQL/MySQL optimization, and API performance tuning, reducing **average response times** by **35%** under peak traffic.
- Contributed to feed, search, and discovery system enhancements, improving content relevance and **user retention** by **15-20%** through smarter ranking algorithms and faster content delivery.

ViSenze | Singapore (Remote) | Oct 2020 - May 2021

AI based Project Engineer - API Integration

- Built RESTful API connectors for enterprise clients to integrate AI-powered visual search capabilities into web and mobile applications, supporting scalable SaaS adoption across **multiple e-commerce platforms** and reducing client integration time by **45%**.

- Assisted in deploying image classification and feature extraction pipelines using Python-based ML frameworks (TensorFlow/PyTorch), improving **image recognition accuracy** and recommendation relevance by **25%**.
- Optimized low-latency image inference services and API processing workflows, reducing average response times by **40%** and improving production stability **under high request volumes**.
- Supported cross-functional teams in translating ML model outputs into production-ready API services for retail recommendation systems

Red Dot Payment (Non-remote) | Nov 2018 - Oct 2020

Full Stack Developer - Backend-heavy SaaS Deployment

- Developed backend services for payment gateway integration and transaction processing systems using **Java** (Spring Boot) and **Node.js**.
- Built **secure APIs** supporting card payments, subscription billing, and reconciliation workflows, aligned with fintech compliance standards.
- Designed **backend logic** for payment status tracking, refund flows, and merchant onboarding systems.
- Worked on deployment pipelines using **Docker** and **AWS** infrastructure, supporting stable production releases for financial transactions.
- Improved **system reliability** by optimizing transaction processing workflows and error-handling mechanisms **under high-load conditions**.

EDUCATION

Singapore Management University (SMU) | 2014 - 2018

Bachelor's Degree in Computer Science | 2014 - 2017

Master's Degree | 2017 - 2018

SKILLS

- **Frontend:** React, TypeScript, Vue.js, TailwindCSS, Shadcn/ui, HTML, CSS, JavaScript
- **Backend & APIs:** Java, Kotlin, Python (Django, FastAPI, PyTorch), Node.js, Go, REST, GraphQL, WebSockets,
- **Databases & Caching:** PostgreSQL, MySQL, Redis, MongoDB
- **Cloud & DevOps:** AWS (EC2, S3, RDS, IAM), Docker, GitHub Actions CI/CD
- **AI & ML Integration:** OpenAI API, LangChain
- **Languages:** English (Native) Japanese (Fluent)